

Course Objectives-To make students familiar with the analysis and design of typical pre-stressed concrete structural elements.as per IS codal provisions.

Introduction-Theory and Basic Concepts: Prestressing concrete terminology, advantages and applications of prestressed concrete, types of prestressing, materials for prestressed concrete and permissible stresses, prestressing systems and devices, losses in prestress.

Limit State Design: Review of limit state design concepts- safety and serviceability requirements.

Analysis and Design for Axial Tension and Flexure: Analysis of members under axial load and flexure, resultant stresses at a section, analysis at service loads-load balancing concept, cracking moment, analysis for ultimate strength, calculation of moment demand, design of type 1 and type 2 members, detailing requirements, analysis of partially prestressed sections.

Analysis and Design for Shear and Torsion: Stress in an uncracked beam, limit state of collapse for shear and torsion, design steps and detailing requirements.

Calculations of Deflection and Crack Width: Total deflection due to gravity load and prestressing force, limits of deflection, limits of span-to-effective depth ratio, prediction of long-term deflections, calculation of crack width, limits of crack width.

Transmission of Prestress in pre-tensioned and post-tensioned members: Introduction, transmission length and end zone reinforcement.

Course learning Outcome (CLO):

Upon completion of this course, the students will be able to:

1. Characterize the materials required and various methods of prestressing and evaluate losses in various pre-stressed members.
2. Analysis of prestressed concrete members under flexure and tension.
3. Evaluate the deflections and crack widths in prestressed concrete members.
4. Design various prestressed concrete structural elements for bending, axial tension, shear and torsion.
5. Analysis and Design of end zone in pre-tensioned and post-tensioned members.

Textbooks:

1. Krishna Raju N., "Prestressed concrete", 5th Edition, Tata McGraw Hill Company, New Delhi,2012.
2. Pandit G. S. and Gupta S. P., "Prestressed Concrete", CBS Publishers and Distributors Pvt. Ltd,2012.
3. Singh S. B., "Analysis and Design of Prestressed Concrete Structures" Willey, 2023.

Reference Material

1. Structural design manual, Precast concrete connection details, Society for the studies in the useof precast concrete, Netherland BetorVerlag, 2009
2. Glower, F., "Structural Pre-cast Concrete", Oxford Publishers (2008)
3. Rajagopalan.N, "Prestressed Concrete", Narosa Publishing House, 2002.

4. Dayaratnam.P., “Prestressed Concrete Structures”, Oxford and IBH, 2013.
5. Lin T.Y. and Ned.H.Burns, “Design of prestressed Concrete Structures”, Third Edition, WileyIndia Pvt. Ltd., New Delhi, 2013.
6. IS1343:2012 Code of Practice for Prestressed Concrete, Bureau of Indian Standards, New Delhi,2012.

EvaluationScheme:

Sr. No.	EvaluationElements	Weightage, (%)
1	MST	30
2	EST	45
3	Sessional(weeklyassessedtutorials)	25